Chapter ML:II (continued)

II. Machine Learning Basics

- Regression
- Concept Learning: Search in Hypothesis Space
- Concept Learning: Search in Version Space
- Evaluating Effectiveness

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Evaluating Effectiveness

Misclassification Rate

Definition 8 (True Misclassification Rate, True Error)

Let X be a feature space with a finite number of elements. Moreover, let C be a set of classes, let $y : X \to C$ be a classifier, and let c be the target concept to be learned. Then the true misclassification rate of y, denoted as $\mathit{Err}^*(y)$, is defined as follows:

$$\mathit{Err}^*(y) = \frac{|\{\mathbf{x} \in X : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|X|}$$

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$$Err^*(y) = \frac{|\{\mathbf{x} \in X : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|X|}$$

Problem:

- Usually the *total function* c is unknown and hence $Err^*(y)$ is unknown.

Solution:

- Based on a set of examples D, estimation of upper and lower bounds for $Err^*(y)$ according to some sampling strategy.

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Remarks:

- Instead of the term “true misclassification rate” we will also use the term “true misclassification error” or simply “true error”.
- The definition of the true misclassification rate implies a unique class assignment for the $\mathbf{x} \in X$, i.e., the absence of label noise.
- The English word “rate” can denote both the mathematical concept of a *flow quantity* (a change of a quantity per time unit) as well as the mathematical concept of a *portion*, a *percentage*, or a *ratio*, which has a stationary (= time-independent) semantics. Note that the latter semantics is meant here when talking about the misclassification rate.
- The German word „Rate“ is often (mis)used to denote the mathematical concept of a portion, a percentage, or a ratio. Taking a precise mathematical standpoint, the correct German words are „Anteil“ or „Quote“. I.e., a semantically correct translation of misclassification rate is „Missklassifikationsanteil“, and not „Missklassifikationsrate“.

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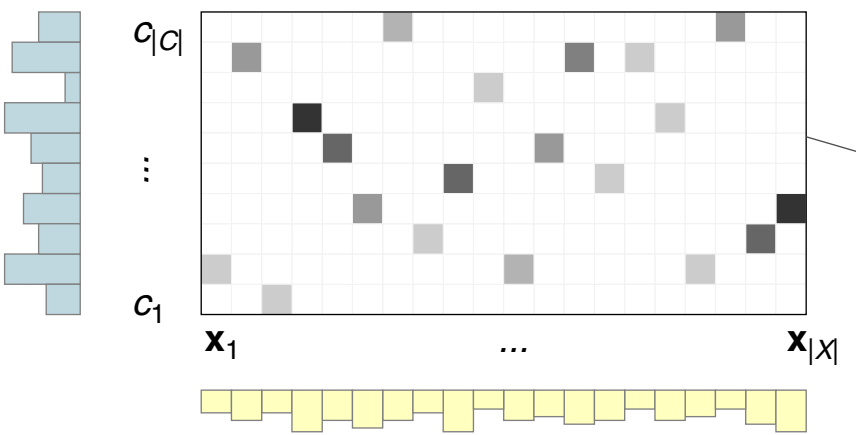
Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation

Let X be a feature space, C a set of classes, and P a probability measure on $X \times C$. Then $P(\mathbf{x}, c)$ (precisely: $P(\mathcal{X} = \mathbf{x}, \mathcal{C} = c)$) denotes the probability of the joint event (1) to observe the vector $\mathbf{x} \in X$ and (2) that $\mathbf{x}$ belongs to class $c \in C$.

Illustration:



Joint probability $P(\mathbf{x}, c)$

(no label noise $\rightarrow$ classes unique)

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

Let X be a feature space, C a set of classes, and P a probability measure on $X \times C$. Then $P(\mathbf{x}, c)$ (precisely: $P(\mathcal{X} = \mathbf{x}, \mathcal{C} = c)$) denotes the probability of the joint event (1) to observe the vector $\mathbf{x} \in X$ and (2) that $\mathbf{x}$ belongs to class $c \in C$.

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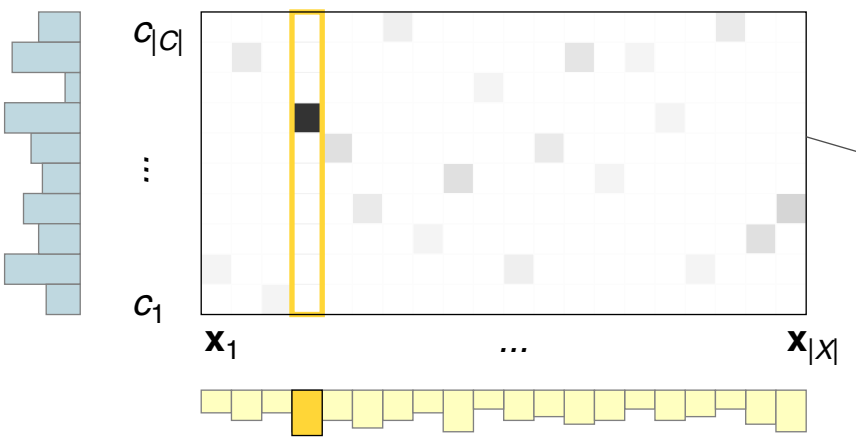

Joint probability $P(\mathbf{x}, c)$

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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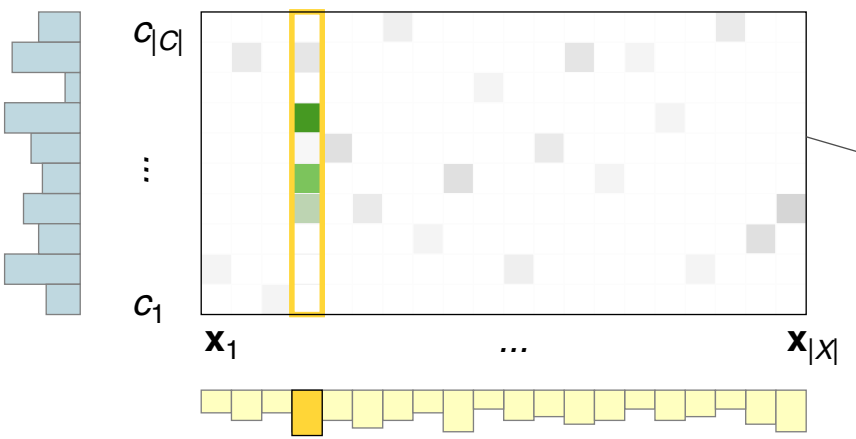

Joint probability $P(\mathbf{x}, c)$

(label noise $\rightarrow$ classes not unique)

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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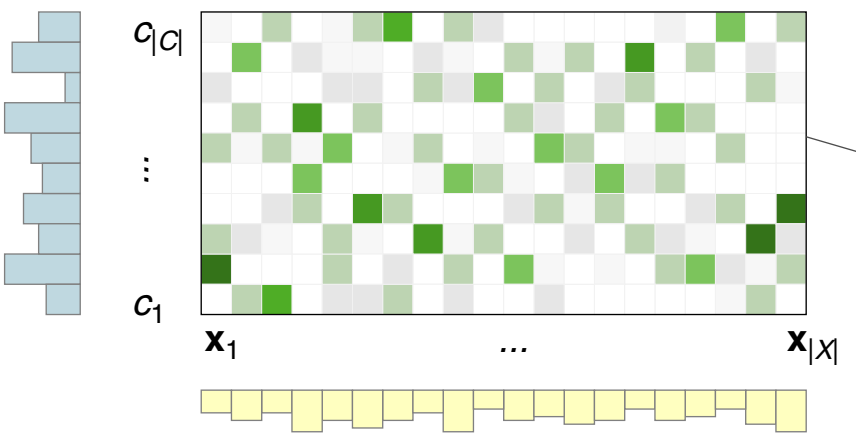

Joint probability $P(\mathbf{x}, c)$

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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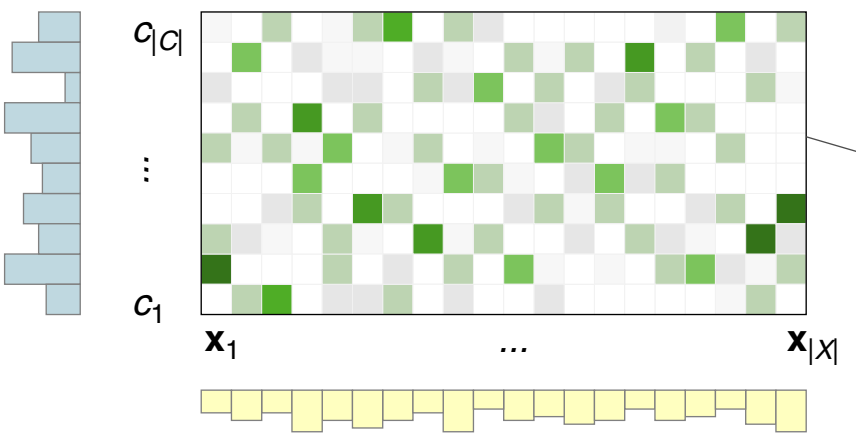

Joint probability $P(\mathbf{x}, c)$

(label noise $\rightarrow$ classes not unique)

$$Err^*(y) = \sum_{\mathbf{x} \in X} \sum_{c \in C} P(\mathbf{x}, c) \cdot I(y(\mathbf{x}), c), \quad \text{with } I(y(\mathbf{x}), c) = \begin{cases} 0 & \text{if } y(\mathbf{x}) = c \\ 1 & \text{otherwise} \end{cases}$$

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Remarks:

- $\mathcal{X}$ and $\mathcal{C}$ denote random variables with domains X and C respectively. In particular, X may not be restricted to a finite set.

- $\mathcal{X}$ models the fact that each observation process is governed by a probability distribution, rendering certain observations more likely than others. Keyword: a-priori probability [of observing some $\mathbf{x}$.]

  Note that the set X in the definition of the true misclassification rate is also governed by this distribution: Accordingly, feature vectors $\mathbf{x}$ that are more likely will occur in X more often and hence are considered with the appropriate weight in $\mathit{Err}^*(y)$.

- $\mathcal{C}$ models the fact that in the real world the classification $c(\mathbf{x})$ of a feature vector $\mathbf{x}$ may not be deterministic. Reasons for a probabilistic class assignment include: an incomplete feature set, imprecision and random errors within the feature measuring process, lack of care during data acquisition. Keyword: label noise.

- P is a probability measure and hence its argument must be an event, such as "$\mathcal{X} = \mathbf{x}$" or "$\mathcal{X} = \mathbf{x} \mid \mathcal{C} = c$". I.e., notations such as $P(\mathbf{x})$ and $P(\mathbf{x} \mid c)$ are abbreviations of $P(\mathcal{X} = \mathbf{x})$ and $P(\mathcal{X} = \mathbf{x} \mid \mathcal{C} = c)$ respectively.

- Let A and B denote two events, e.g., $A =$ "$\mathbf{x}$ [was observed]" and $B =$ "c [is the class of $\mathbf{x}$]". Then the following expressions are syntactic variants to denote the probability of the combined event: $P(A, B)$, $P(A \text{ and } B)$, $P(A \wedge B)$.

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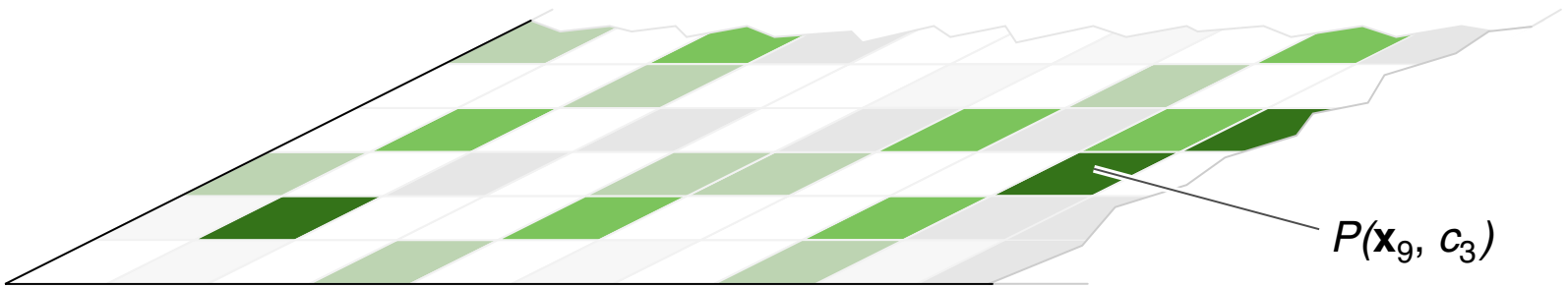
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions

Joint probability distribution $P(\mathbf{x}, c)$:



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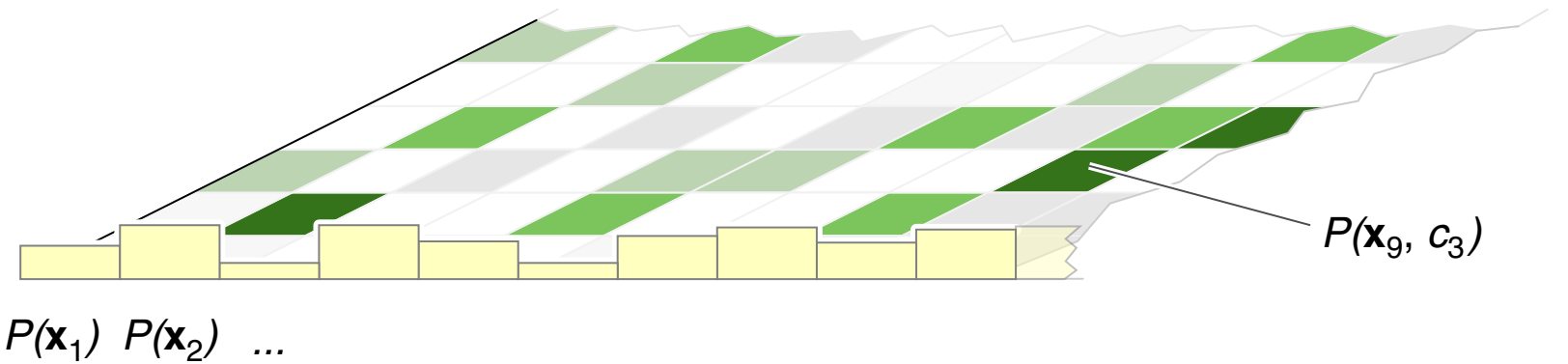
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions (continued)

Marginal probability distribution $P(\mathbf{x})$:



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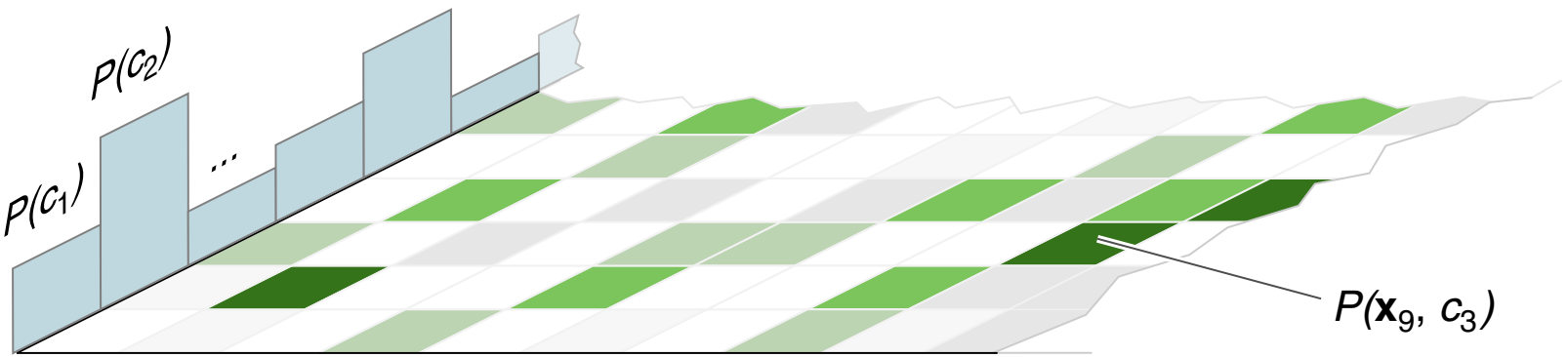
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions (continued)

Marginal probability distribution $P(c)$:



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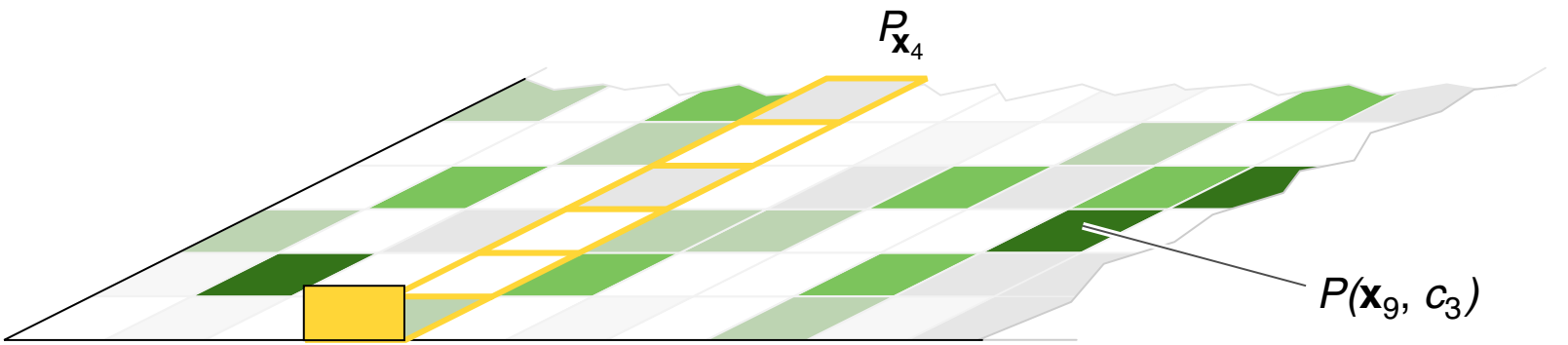
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions (continued)

Probability distribution $P_{\mathbf{x}}(c)$ or $P_{\mathcal{X}=\mathbf{x}}(c)$ under feature vector $\mathbf{x}$:



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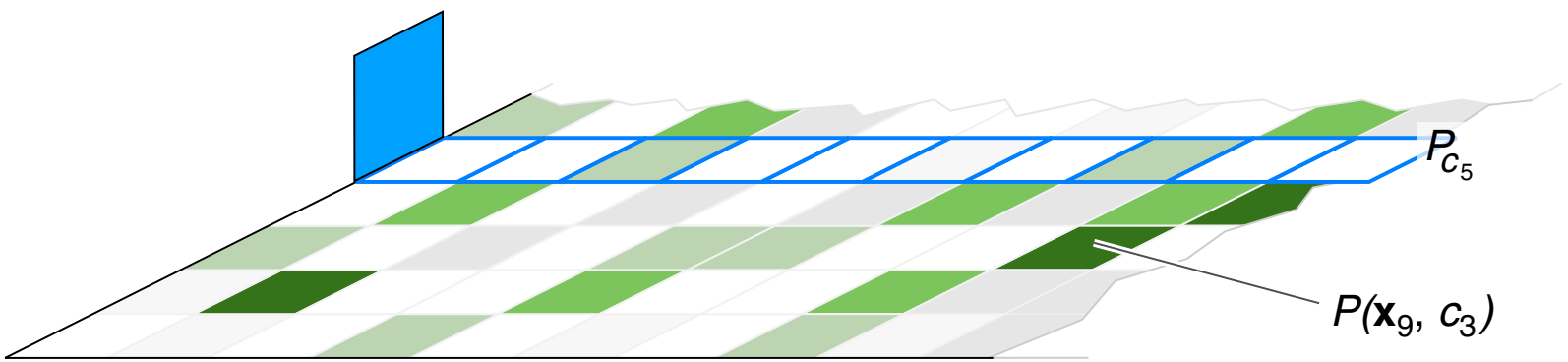
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions (continued)

Probability distribution $P_c(\mathbf{x})$ or $P_{\mathcal{C}=c}(\mathbf{x})$ under class c:



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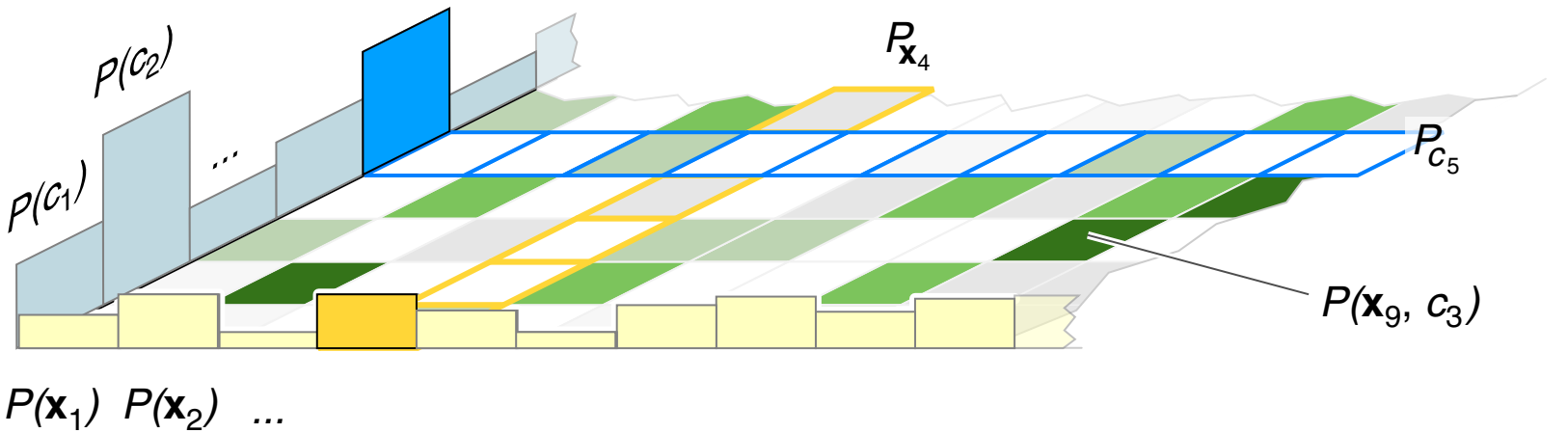
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions (continued)

Overview:



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Remarks:

- $P_{\mathbf{x}}(c) \equiv P(c \mid \mathbf{x})$. $P_{\mathbf{x}}$ is a probability measure, also called "conditional class probability" function.
  In the illustration: $P_{\mathbf{x}_4}$ is the distribution of the c (consider the column) for feature vector $\mathbf{x}_4$.
  Summation over the $c \in C$ in yields the marginal probability $P(\mathbf{x}_4)$.

- $P_c(\mathbf{x}) \equiv P(\mathbf{x} \mid c)$. P_c is a probability measure, also called "class-conditional probability [density]" function.
  In the illustration: P_{c_5} is the distribution of the $\mathbf{x}$ (consider the row) for class c_5.
  Summation / integration over the $\mathbf{x} \in X$ yields the marginal probability $P(c_5)$.

- $P(\mathbf{x}, c) = P(\mathbf{x} \mid c) \cdot P(c)$, where $P(c)$ is the a-priori probability for (observing) event c, and $P(\mathbf{x} \mid c)$ is the probability for (observing) event $\mathbf{x}$ given event c.
  Likewise, $P(\mathbf{x}, c) = P(c, \mathbf{x}) = P(c \mid \mathbf{x}) \cdot P(\mathbf{x})$, where $P(\mathbf{x})$ is the a-priori probability for (observing) event $\mathbf{x}$, and $P(c \mid \mathbf{x})$ is the probability for (observing) event c given event $\mathbf{x}$.

- Let both events $\mathbf{x}$ and c have occurred already, and, let $\mathbf{x}$ be known and c be unknown. Then, $P(\mathbf{x} \mid c)$ is called *likelihood* (for event $\mathbf{x}$ given event c).

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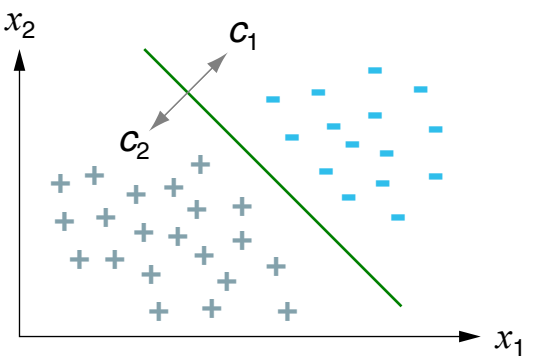
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes



$$X = \left\{ \begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}, \begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}, \begin{pmatrix} x_{3_1} \\ x_{3_2} \end{pmatrix}, \ldots \right\}$$

$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \mathbf{x}_3 \qquad \ldots$$

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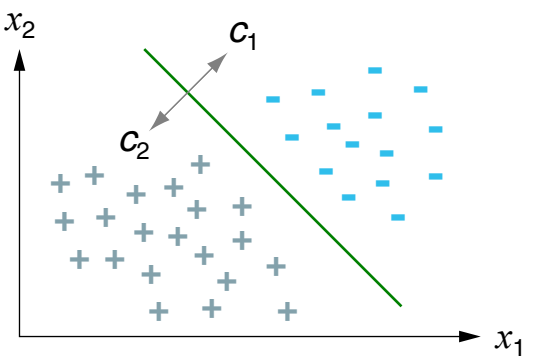
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



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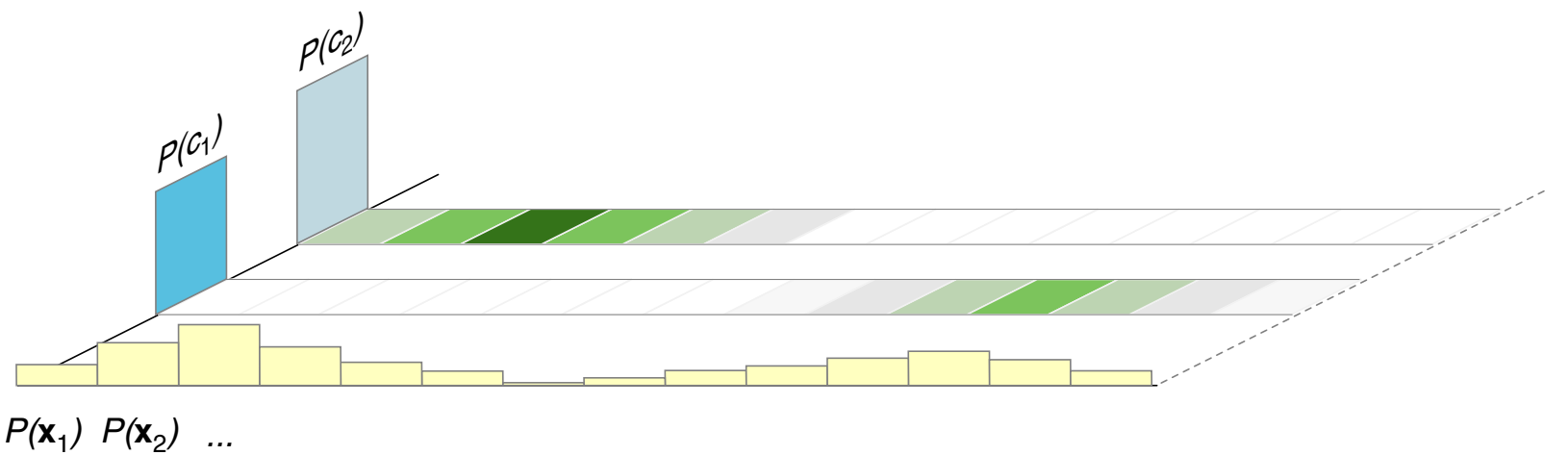
$$\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \ldots$$

Joint probability distribution $P(\mathbf{x}, c)$, marginal distributions $P(\mathbf{x})$ and $P(c)$:



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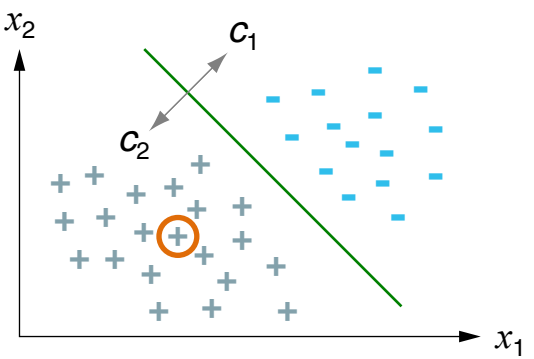
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



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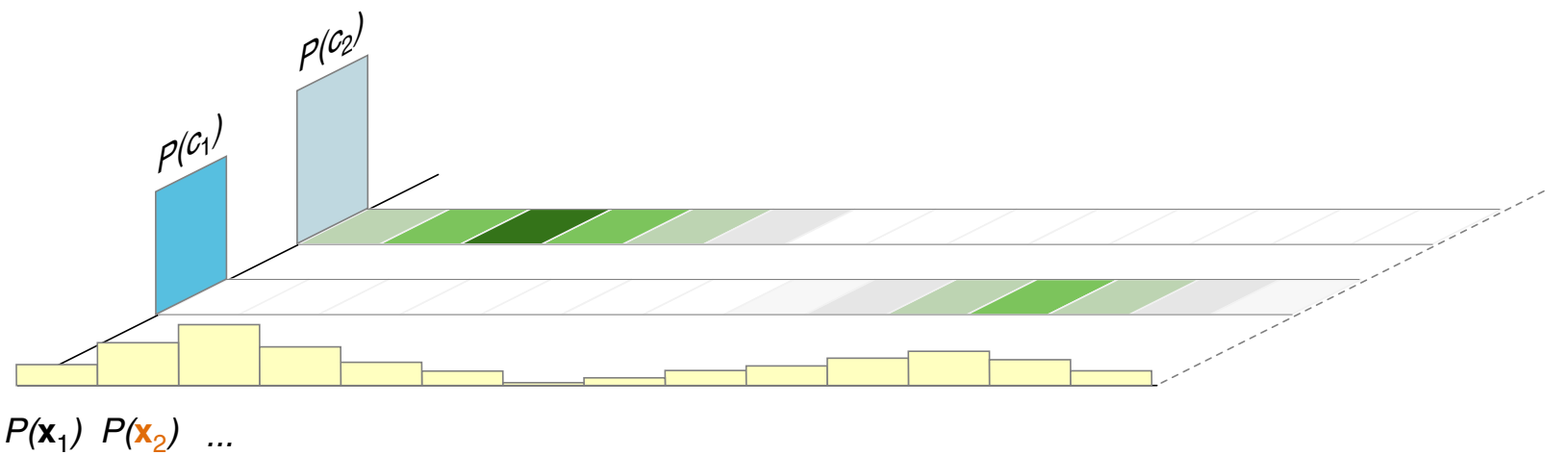

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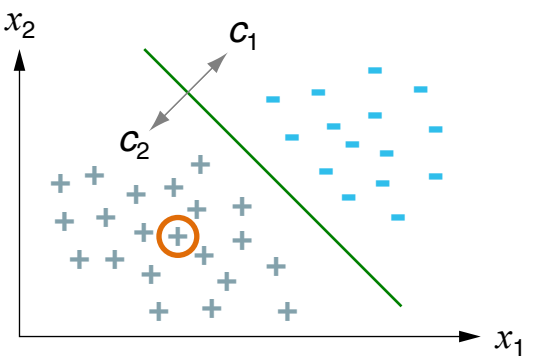
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



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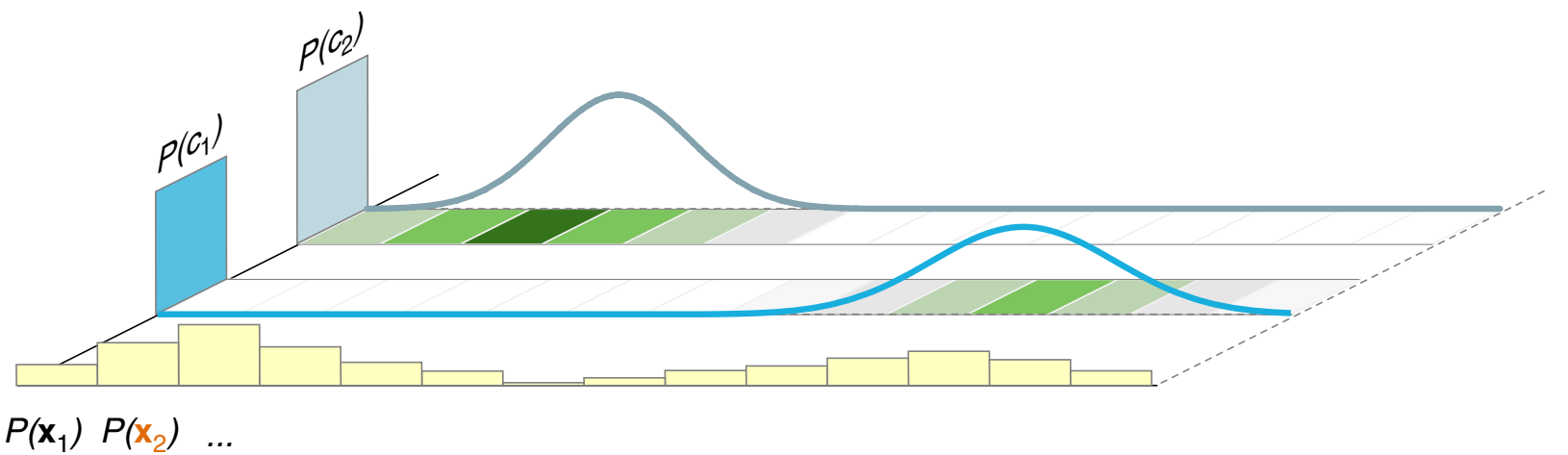
$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \mathbf{x}_3 \qquad \ldots$$

Class-conditional probability distributions $P(\mathbf{x} \mid c_1)$ and $P(\mathbf{x} \mid c_2)$:



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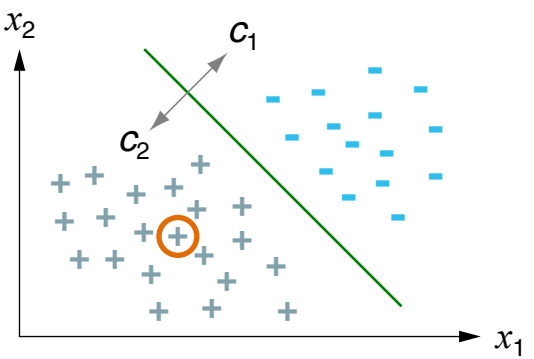
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



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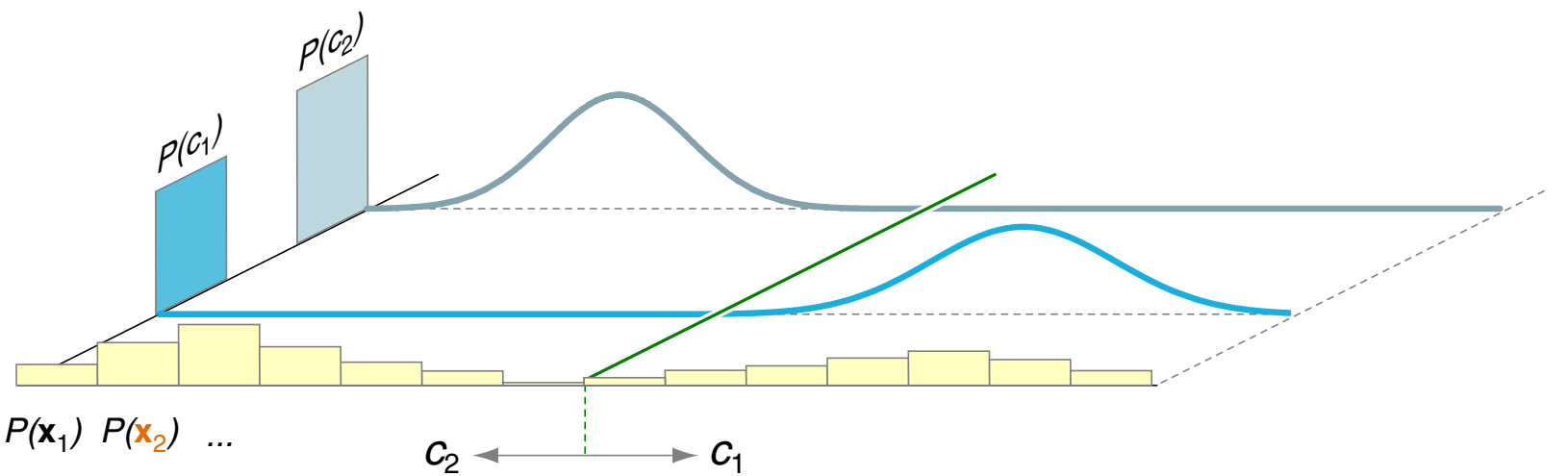
Effect of a hyperplane classifier:



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Remarks:

- Linear separability requires a feature space with a particular structure. Here we assume that the feature space is a vector space over the scalar field of real numbers $\mathbf{R}$, equipped with the dot product.
- Recall that in the definition of the true misclassification rate nothing is presumed about the structure of the set X.

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Evaluating Effectiveness

Estimating Error Bounds

Classifier training:

- $D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times C$ is a set of examples.
- $y : X \to C$ is the classifier trained on D.
- The true error $Err^*(y)$ measures the performance of y on X ("in the wild").
- What can be said about the true error $Err^*(y)$ of y?

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Evaluating Effectiveness

Estimating Error Bounds

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The following relations typically hold:

Training error		Cross-validation error		Holdout error
$Err_{tr}(y)$	$<$	$Err_{cv}(y, k)$	$<$	$Err(y, D_{test})$

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Evaluating Effectiveness

Estimating Error Bounds

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- What can be said about the true error $Err^*(y)$ of y?

The following relations typically hold:

Training error		Cross-validation error		Holdout error		True error
$Err_{tr}(y)$	$<$	$Err_{cv}(y, k)$	$<$	$Err(y, D_{test})$	$<$	
	$\leftarrow$					$Err^*(y)$

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Remarks:

- Relating the true error $Err^*(y)$ to the aforementioned error assessments $Err_{tr}(y)$, $Err_{cv}(y,k)$, and $Err(y, D_{test})$ is not straightforward but requires an in-depth analysis of the sampling strategy, the sample size D, and the feature space X, among others.
- With $D = X$ the true error $Err^*(y)$ becomes the training error $Err_{tr}(y)$.

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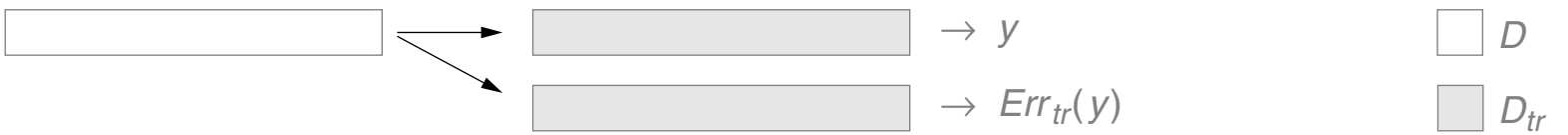
Evaluating Effectiveness

(1) Training Error



Setting:

- No test set.
- y is the classifier trained on $D_{tr} = D$.

Training error of y:

- $$Err_{tr}(y) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{tr} : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{tr}|}$$

  $=$ misclassification rate of y on the training set.

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Remarks:

- $\mathit{Err}_{tr}(y)$ is based on examples that are also exploited to train y. It hence quantifies the *memorization* capability but not the *generalization* capability of y.
- $\mathit{Err}_{tr}(y)$ is an optimistic estimation, i.e., it is constantly lower compared to the (unknown) true error $\mathit{Err}^*(y)$.
- Note that the above issues relate to the meaningfulness of $\mathit{Err}_{tr}(y)$ as an error estimate—not to the classifier y. Obviously, to get the maximum out of the data when training y, D must be exploited completely: A classifier y trained on D will on average outperform any classifier y' trained on a subset of D. Hence, the performance of y' is an upper bound for the (unknown) performance of y.
- Regarding notation: We will us the prime symbol »'« to indicate whether a classifier is trained by withholding a test set. Examples: y', y'_i

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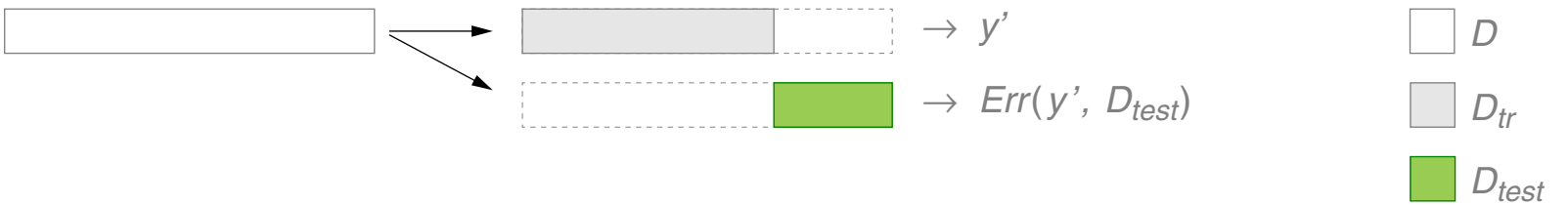
Evaluating Effectiveness

(2a) Holdout Error



Setting:

- $D_{test} \subset D$ is the test set.
- y' is the classifier trained on $D_{tr} = D \setminus D_{test}$.

Holdout error of y :

- $$Err(y, D_{test}) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{test} : y'(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{test}|}$$

  $=$ misclassification rate of y' on the test set.

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Remarks:

- When splitting D into D_{tr} and D_{test} one has to ensure that the underlying distribution is maintained, i.e., the examples have to be drawn independently and according to P. If this condition is not fulfilled then $Err(y, D_{test})$ cannot be used as an estimation of $Err^*(y)$.
  Keywords: stratification, sample selection bias

- D_{tr} and D_{test} should have similar sizes. A typical value for splitting D into training set D_{tr} and test set D_{test} is 2:1.

- The fact that random variables are both independent of each other and identically distributed is abbreviated with "i.i.d."

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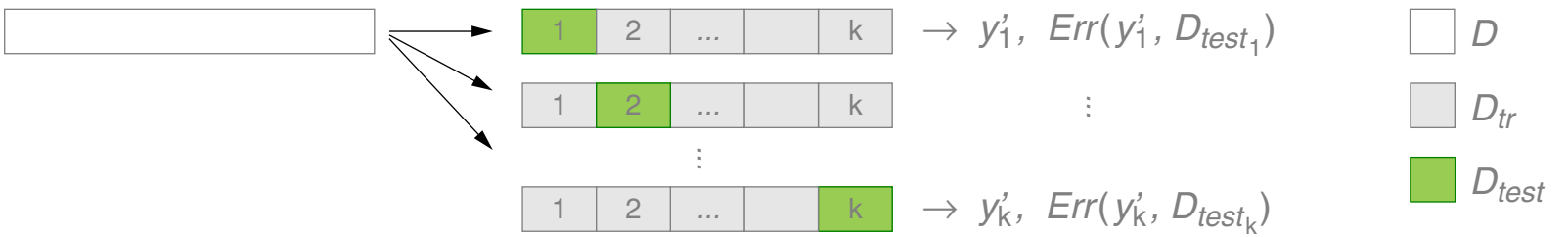
Evaluating Effectiveness

(2b) k-Fold Cross-Validation



Setting:

- k test sets D_{test_i} by splitting D into k disjoint sets of similar size.

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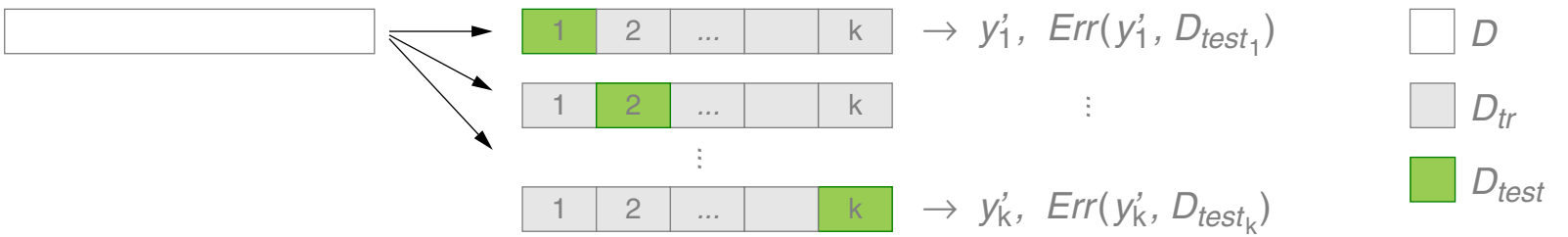
Evaluating Effectiveness

(2b) k-Fold Cross-Validation



Setting:

- k test sets D_{test_i} by splitting D into k disjoint sets of similar size.
- $i = 1, \ldots, k$:

  y'_i is the classifier trained on $D_{tr} = D \setminus D_{test_i}$,

$$Err(y'_i, D_{test_i}) = \frac{|\{(\mathbf{x}, c(\mathbf{x}) \in D_{test_i} : y'_i(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{test_i}|}$$

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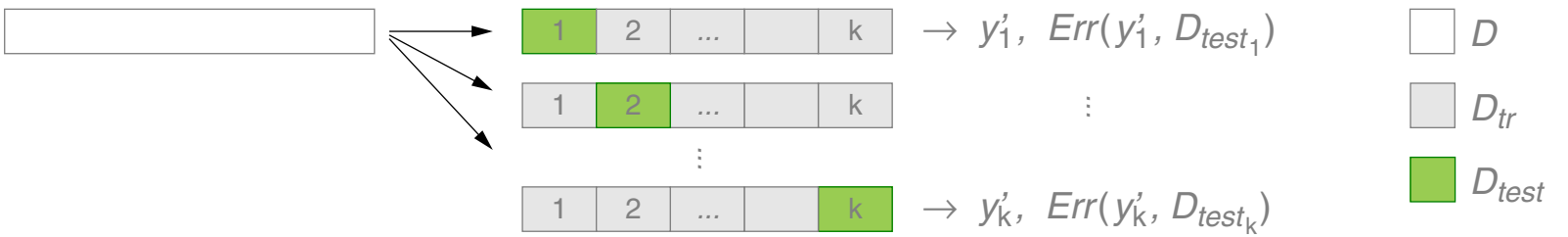
Evaluating Effectiveness

(2b) k-Fold Cross-Validation



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Cross-validation error of y :

- $$Err_{cv}(y, k) = \frac{1}{k} \sum_{i=1}^{k} Err(y'_i, D_{test_i})$$

  $=$ averaged misclassification rate on the k test sets.

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Remarks:

- For large k the set $D_{tr} = D \setminus D_{test_i}$ is of similar size as D. Hence $Err(y_i, D_{test_i})$—as well as $Err_{cv}(y, k)$—is close to $Err^*(y)$, since $Err^*(y)$ is the error of the classifier y trained on D.
- n-fold cross-validation (aka "leave one out") is the special case with $k = n$. Obviously singleton test sets ($|D_{test_i}| = 1$) are never stratified since they contain a single class only.
- n-fold cross-validation is a special case of exhaustive cross-validation methods, which learn and test on all possible ways to divide the original sample into a training and a validation set.

[Wikipedia]

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Evaluating Effectiveness

Comparing Model Variants

Classifier training:

- $D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times C$ is a set of examples.
- m hyperparameter values $\pi_1, \pi_2, \ldots, \pi_m$,

  m classifiers $y_{\pi_1}, y_{\pi_2}, \ldots, y_{\pi_m}$.
- $y_{\pi_j} : X \to C$ is the classifier trained on D.
- Which is the most effective among the m classifiers y_{π_j}?

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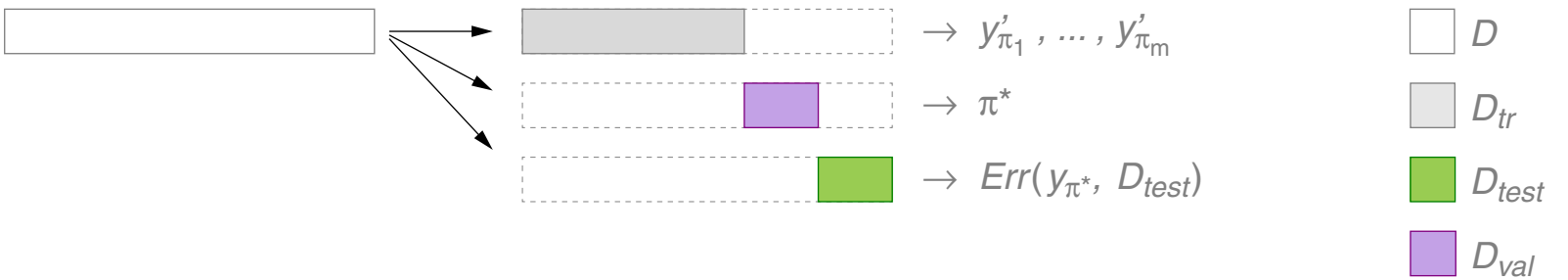
Evaluating Effectiveness

(3a) Model Selection (single validation set)



Setting:

- $D_{test} \subset D$ is the test set,

  $D_{val} \subset (D \setminus D_{test})$ is the validation set.

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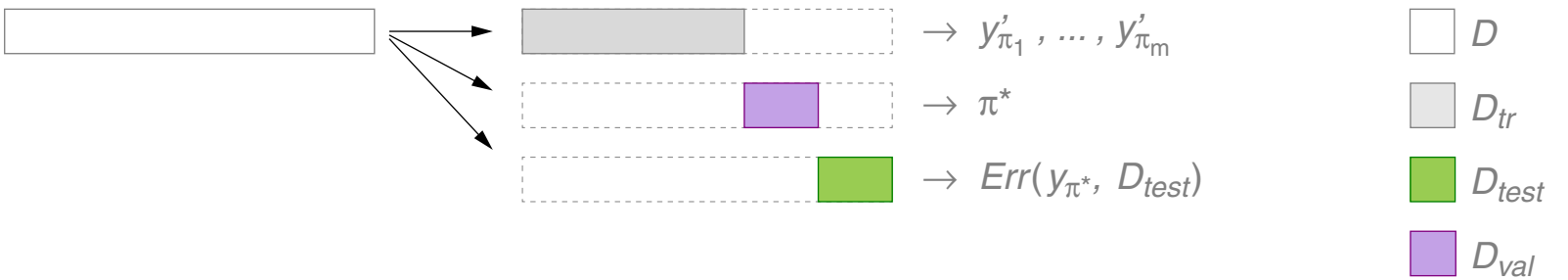
Evaluating Effectiveness

(3a) Model Selection (single validation set)



Setting:

- $D_{test} \subset D$ is the test set,

  $D_{val} \subset (D \setminus D_{test})$ is the validation set.

- $j = 1, \ldots, m$:

  y'_{π_j} is the classifier trained on $D_{tr} = D \setminus (D_{test} \cup D_{val})$,

$$Err(y'_{\pi_j}, D_{val}) = \frac{|\{(\mathbf{x}, c(\mathbf{x}) \in D_{val} : y'_{\pi_j}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{val}|}$$

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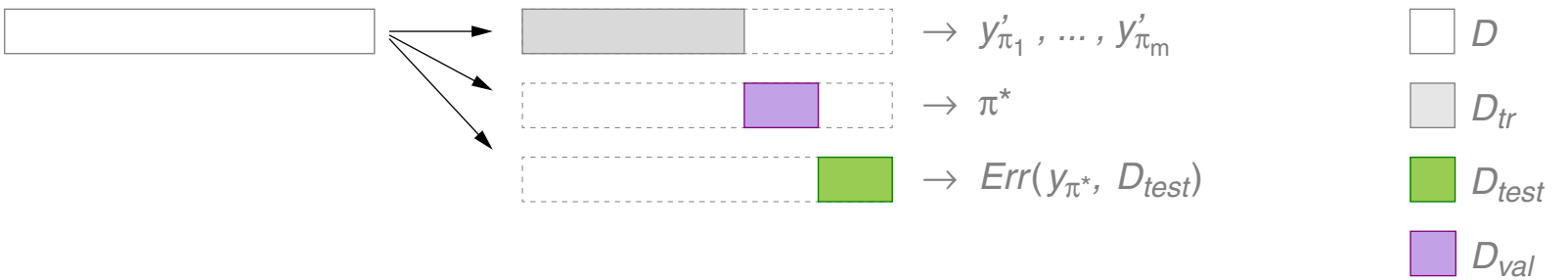
Evaluating Effectiveness

(3a) Model Selection (single validation set) [Holdout Error]



(continued)

- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \ Err(y'_{\pi_j}, D_{val})$
- y_{π^*} is the classifier trained on D,

  y'_{π^*} is the classifier trained on $D_{tr} = D \setminus D_{test}$.

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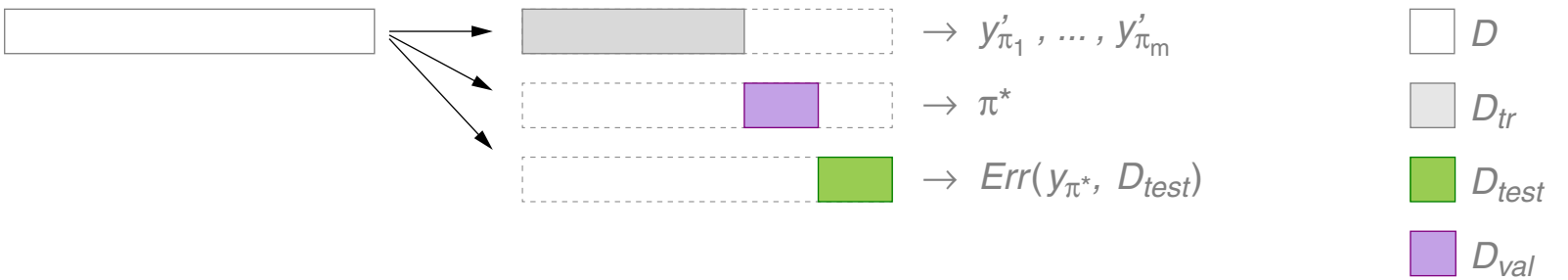
Evaluating Effectiveness

(3a) Model Selection (single validation set) [Holdout Error]



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- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \ Err(y'_{\pi_j}, D_{val})$
- y_{π^*} is the classifier trained on D,

  y'_{π^*} is the classifier trained on $D_{tr} = D \setminus D_{test}$.

Holdout error of y_{π^*} :

- $Err(y_{\pi^*}, D_{test}) = \dfrac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{test} : y'_{\pi^*}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{test}|}$

  $=$ misclassification rate of y'_{π^*} on the test set.

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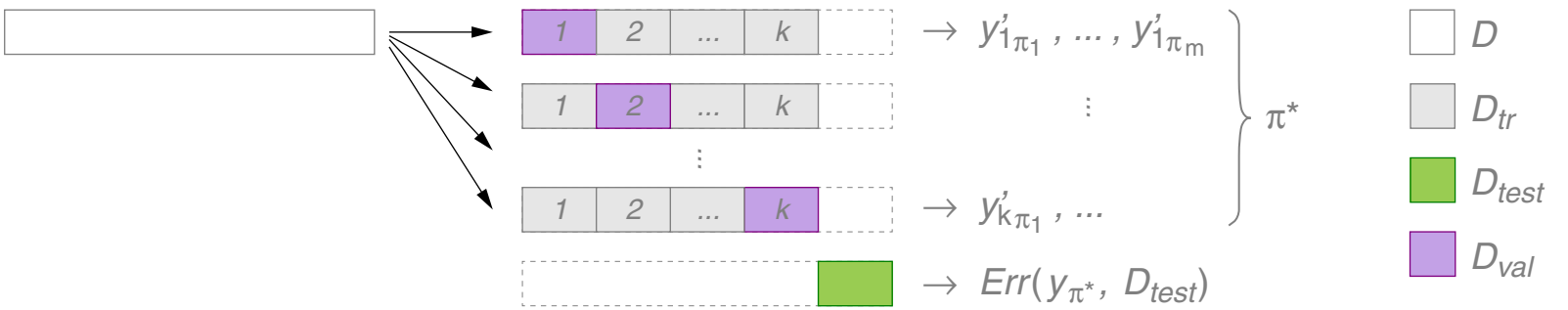
Evaluating Effectiveness

(3b) Model Selection (k validation sets)



Setting:

- $D_{test} \subset D$ is the test set,

  k validation sets D_{val_i} by splitting $D \setminus D_{test}$ into k disjoint sets of similar size.

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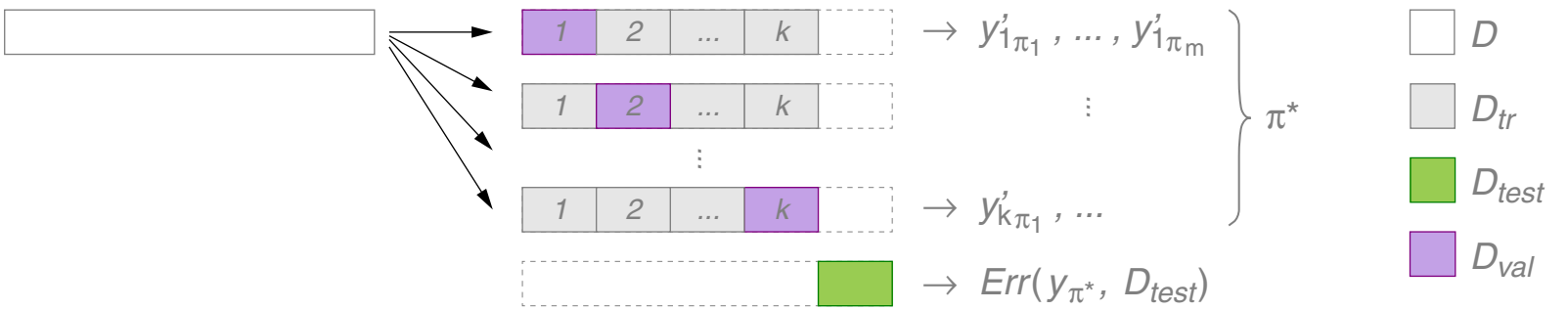
Evaluating Effectiveness

(3b) Model Selection (k validation sets)



Setting:

- $D_{test} \subset D$ is the test set,

  k validation sets D_{val_i} by splitting $D \setminus D_{test}$ into k disjoint sets of similar size.

- $i = 1, \ldots, k$:

  $j = 1, \ldots, m$:

  $y'_{i_{\pi_j}}$ is the classifier trained on $D_{tr} = D \setminus (D_{test} \cup D_{val_i})$,

$$Err(y'_{i_{\pi_j}}, D_{val_i}) = \frac{|\{(\mathbf{x}, c(\mathbf{x}) \in D_{val_i} : y'_{i_{\pi_j}}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{val_i}|}$$

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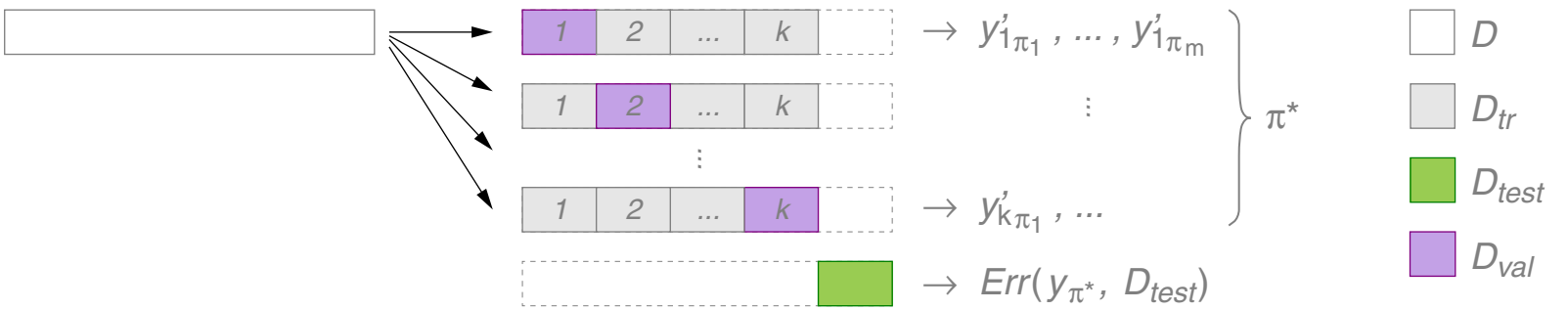
Evaluating Effectiveness

(3b) Model Selection (k validation sets)



(continued)

- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \sum_{i=1}^{k} Err(y'_{i_{\pi_j}}, D_{val_i})$

- y_{π^*} is the classifier trained on D,

  y'_{π^*} is the classifier trained on $D_{tr} = D \setminus D_{test}$.

Holdout error of y : computation as before.

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Remarks:

- The validation set is also called “development set”.

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Evaluating Effectiveness

Misclassification Costs

Use of a *cost measure* for the misclassification of a feature vector $\mathbf{x} \in X$ in a wrong class c' instead of in the correct class c:

$$\mathit{cost}(c' \mid c) \begin{cases} \geq 0 & \text{if } c' \neq c \\ = 0 & \text{otherwise} \end{cases}$$

Holdout error of y based on misclassification costs:

$$\mathit{Err}_{\mathit{cost}}(y, D_{\mathit{test}}) = \frac{1}{|D_{\mathit{test}}|} \cdot \sum_{(\mathbf{x}, c(\mathbf{x})) \in D_{\mathit{test}}} \mathit{cost}(y(\mathbf{x}) \mid c(\mathbf{x}))$$

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Remarks:

- The true error, $Err^*(y)$, is a special case of $Err_{cost}(y)$ with $cost(c' \mid c) = 1$ for $c' \neq c$. Consider in this regard the notation of $Err^*(y)$ in terms of the function $I(y(\mathbf{x}), c)$:

$$Err^*(y) = \frac{|\{\mathbf{x} \in X : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|X|} = \sum_{\mathbf{x} \in X} I(y(\mathbf{x}), c)$$

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